LICET – FNH –STUDY MATERIALS
UNIT III NANOTECHNOLOGY AND FOOD ADDITIVES

The field of nanotechnology, which is expanding quickly, deals with the manipulation and control of matter at the nanoscale, or the scale of atoms and molecules. It entails creating and engineering products with special qualities and features that are not present in their bulk counterparts. Nanotechnology is being used in many different industries, and one area where it shows significant potential is in maintaining food safety.
Consumers, regulators, and the food industry as a whole are very concerned about food safety. Serious health concerns can result from food contamination, deterioration, or the presence of toxic bacteria or chemicals. Enhanced food safety procedures, better packaging materials, and novel techniques for detecting and eradicating contaminants are just a few of the creative responses that nanotechnology offers to these problems.

Definition of Nanomaterials
Materials with at least one dimension in the nanoscale range, typically between 1 and 100 nanometers, are referred to as nanomaterials. This includes many different types of structures, such as nanoparticles, nanofibers, nanowires, nanotubes, and nanosheets. Nanomaterials' distinct features and functionalities are influenced by their dimensionality, composition, and surface qualities.

Unique Properties and Characteristics at the Nanoscale
Unique characteristics and phenomena that are not seen in bulk materials are introduced at the nanoscale. Among the essential traits of nanomaterials are:

1.Increased surface area: When compared to bulk materials, nanomaterials have a much higher surface-to-volume ratio. The reactivity, adsorption capacity, and contact with the environment are all improved by the increased surface area.

2.Quantum effects: Quantum confinement effects become more pronounced at the nanoscale and have an impact on the electrical and optical characteristics of nanomaterials. New capabilities and applications can be made possible by quantum effects that result in size-dependent changes in conductivity, bandgap, and optical characteristics.

3. **Mechanical properties**: Nanomaterials frequently have extraordinary mechanical flexibility and strength. They are appropriate for reinforcing and structural applications because they can have high tensile strength, resilience, and hardness.

4. **Enhanced thermal properties**: Compared to bulk materials, nanomaterials can have better thermal conductivity and heat dissipation. Because of this, they are useful as materials for thermal interfaces, energy conversion, and heat control.

5. **Tailorable properties**: By carefully regulating a nanomaterial's size, shape, composition, and surface changes, its properties can be precisely tailored. The design of materials with the desired qualities for particular applications is made possible by the capacity to adjust properties.

Classification of nanomaterials based on dimension

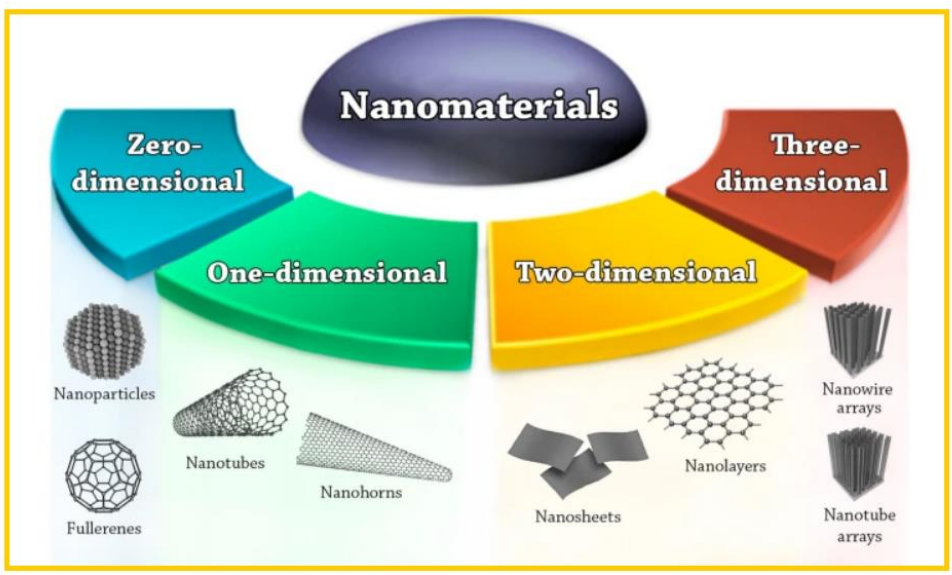
One way to classify nanomaterials is by the number of dimensions not confined to the nanoscale range. Zero-dimensional nanomaterials have all dimensions within the nanoscale zero dimensions (0D), but no dimensions are larger than 100 nm. They are the most common nanoparticles. One-dimensional nanomaterials have one dimension (1D) outside the nanoscale. They include nanotubes, nanorods, and nanowires. Two-dimensional nanomaterials have two dimensions (2D) not confined to the nanoscale. They include nanofilms, nanolayers, and nanocoatings. Three-dimensional nanomaterials, or bulk nanomaterials have all three dimensions (3D) outside the nanoscale.

Nanomaterial types

Carbon-based nanomaterial types are composed primarily of carbon and have the form of ellipsoids, hollow spheres and tubes. Metal-based nanomaterials include nanosilver, nanogold, quantum dots, and nano oxides like titanium dioxide. Dendrimers are nanosized polymers made from branched units. And composites combine nanoparticles with different nanoparticles or with bigger and more bulky materials.

2D materials

One of the first 2D materials to trigger further research on these kinds of materials is graphene. Many studies regarding energy conversion and storage have looked at the fascinating properties of 2D nanomaterials. But there is more: Metal layers, e.g., play an important role in the visual performance of the display assembly. Nanocrystalline TiO2 coatings are among the most common materials for hard tissue replacement, as they provide suitable mechanical and chemical properties. Self-assembled monolayers (SAMs) are spontaneously formed assemblies of organic molecules adsorbed on solid surfaces often used for biosensors.



Nanomaterials classification based on dimensionality

Quantum dots
Quantum dots are artificial semiconducting crystals in the nanoscale which can transport electrons. When UV light hits them, they can emit light in various colors. One way to produce these materials is microwave-assisted colloidal synthesis. Their uses include solar cells, biological applications, LED displays, photodetectors, photocatalysts etc.

Nanowires, nanofibres, nanorods
These materials have two dimensions in the nanoscale and are produced through synthesis. Whereas nanowires are used for different kinds of field-effect transistors, sensors, lasers, and more, the most important nanofiber applications are tissue engineering, drug delivery, cancer diagnosis, optical sensors, air filtration, textiles, etc. Nanorods are solid nanofibers and are important due to their superior performance in energy storage.

Carbon nanotubes
Carbon nanotubes (CNTs) are allotropes of carbon with a cylindrical nanostructure. They are composed of one (single-walled CNT) or several (multi-walled CNT) layers of carbon with a graphitic structure wrapped around a hollow core. Both the dimensions of this core and the wall are in the range of nanometers, while the overall length of the tubes is typically much longer. Composites of CNTs dispersed in matrix materials (e. g. polymers) show very interesting, novel properties. This makes them potentially useful in many fields, such as materials science, electronics, optics and others.

Biological nanomaterials
Biological nanomaterials can have almost all shapes described above, but mostly come into play as nanoparticles for drug delivery in the form of exosomes or regenerative medicine or as inactivated virus and virus-like particles in vaccines. Polymeric drug coatings (conventionally used as a strategy for controlled delivery and enhanced stability) also belong to biological nanomaterials.

Suspensions and dispersions
Nanoparticle dispersions are suspensions of nanoparticles in liquids – water or organic solvents. Such dispersions can be used in different ways: as-is or diluted with appropriate solvents. Suspensions of nanoparticles are of high importance in e.g. the chemical and biological industries, material science, and medicine. Typical application areas, requiring homogenous dispersion, are paints and coatings, inks, drug delivery, and ceramic and nanocomposite processing.

Composites
Composite nanomaterials have a composite structure developed by two or more components of nanoscale, presenting extraordinary physical and chemical properties. Thanks to these properties, along with their composition and chemical and structural flexibility, nanocomposites are not only highly relevant for fundamental research areas, but also for use in numerous novel application fields. Nanocomposites have gained significance in various industries – ranging from small-scale to huge manufacturing sites. They have enormous potential to support the development of advanced and environmentally friendly technologies in industrial sectors like the automotive sector, construction, and electronics.

Membranes

Nanomaterial-based membranes are often the basis for the design and development of novel applications. They possess outstanding properties – e.g. great physical and chemical stability – making them an important resource in research and for many industrial applications.

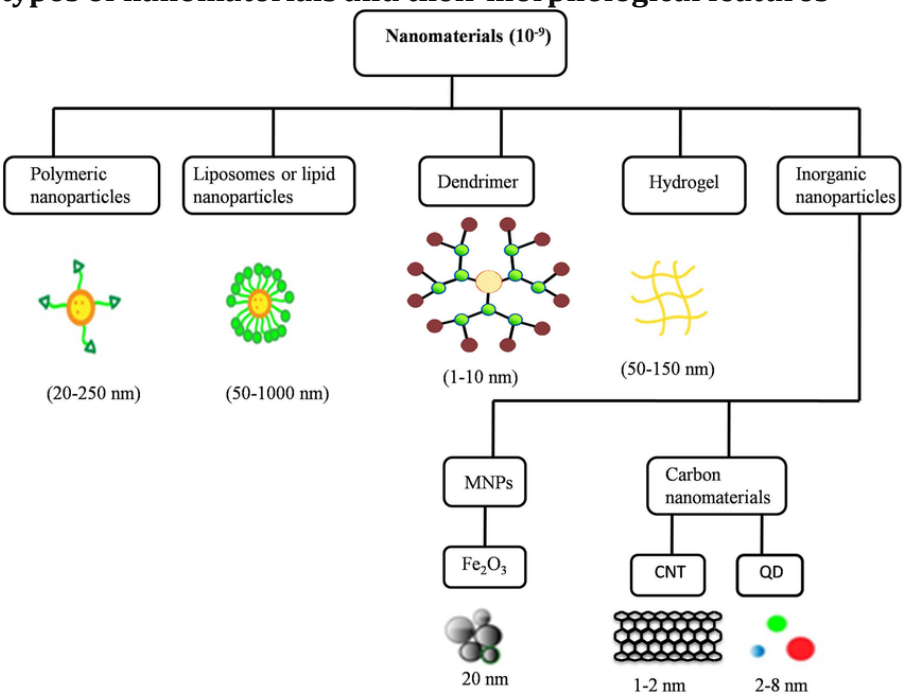
Polymers

Polymers are firmly established in technological advancements and in our daily lives due to their excellent chemical and physical properties, processability and relatively low cost. Polymers are also called macromolecules, which are simply big molecules. These are long chains comprised of small molecules bonded together. A deep understanding of polymers and the development of polymer properties in polymeric nanotechnology provide a broad array of application opportunities e.g. in the pharmaceutical and biotechnology industries.

Semiconductors

Reducing semiconductor materials to nanoscale radically changes their physical and chemical properties, resulting in exceptional properties based on their large surface area and quantum size effect, nonlinear optical properties and luminescence. Semiconductor nanomaterials, e.g. semiconductor nanoparticles, have attracted a lot of attention in the past few years because of the research promise they hold for applications in many different fields. A few examples are information processing, solar cells, nanoscale electronic devices, and new sensor technologies like biosensors. Further steps in these developmental advancements of nanotechnology will initiate substantial breakthroughs in the semiconductor industry.

Fig.Various types of nanomaterials and their morphological features



Importance and Applications in Various Industries

Nanomaterials have become quite important and have applications in many different industries, such as:

1. **Electronics and Optics**: Electronic component shrinking, high-density data storage, and the creation of nanoscale devices are all made possible by nanomaterials. They are utilized for lighting and imaging purposes in displays, sensors, transistors, solar cells, and quantum dots.

2. **Medicine and Biotechnology**: Drug delivery methods, biological imaging, and tissue engineering all rely heavily on nanomaterials. They provide more precise therapy delivery, advance diagnostic methods, and offer tissue regeneration scaffolds.

3. **Energy and Environment**: In order to improve the performance and capacity of energy storage devices like batteries and supercapacitors, nanomaterials are used. For effective energy conversion, they are also utilized in solar cells, fuel cells, and catalysis. Nanomaterials are useful in environmental applications for air filtration, water purification, and environmental cleanup.

4. **Automotive and Aerospace**: The lightweight, high-strength materials utilized in coatings, airplane constructions, and car parts are made possible by nanomaterials. Their features improve durability, performance, and fuel economy while having a smaller negative environmental impact.

5. **Consumer Products**: Everyday products, such as cosmetics, textiles, food packaging, and home appliances, are increasingly incorporating nanomaterials. In these consumer goods, they provide better functionality, toughness, and safety.

With current research continuing to open up new opportunities, the wide range of uses of nanomaterials illustrates their enormous potential for innovation and advancement across many industries. But for nanomaterials to be used safely and sustainably, responsible development, evaluation of possible dangers, and ethical concerns are essential.

Nanotechnology in Food Safety:

1. **Food Packaging**: The creation of food packaging materials has been completely transformed by nanotechnology. Nanocomposite films and coatings can be created with barrier qualities that stop gases, moisture, and other pollutants from migrating through them. This aids in prolonging the shelf life of food as well as preserving its quality and freshness for longer
periods of time. Additionally, dangerous gases or pathogens can be detected and signaled by nanosensors integrated into packaging, creating an early warning system for possible contamination.

2. **Foodborne Pathogen Detection**: Conventional techniques for identifying foodborne pathogens are frequently labor-intensive and call for a laboratory setting. Rapid and sensitive detection methods made possible by nanotechnology can more effectively identify pathogens including

bacteria, viruses, and parasites. Food samples can be quickly screened using nanoscale biosensors and nanomaterial-based assays, cutting down on analysis time and enhancing food safety.

3. **Removal of Pollutants from Food and Water**: Nanomaterials have special features that make them efficient at doing this. For instance, nanoparticles can be created to attach just to certain contaminants, like heavy metals, pesticides, or mycotoxins, and then remove them from food matrixes. The procedure, referred to as "nanosorbent-based remediation," offers a focused and effective way to lower the amounts of dangerous compounds in food.

4. **Nutrient Delivery and Enhancement**: Nanotechnology also contributes to the enhancement of food's nutritional content and delivery. Vitamins, antioxidants, and probiotics can all be enclosed in nanoparticles or nanostructured materials through the process of nanoencapsulation. As a result, their stability, bioavailability, and controlled release are increased, which improves nutrient absorption and boosts customers' health.

5. **Risk Assessment and Regulation**: The quick development and deployment of nanotechnology in the food sector has sparked worries about its possible effects on the environment and human health. Because of this, regulatory bodies from all over the world are actively interested in evaluating the safety of nanomaterials used in food products. To ensure consumer safety, efforts are being made to create rules and regulations for the responsible application of nanotechnology in food production.

When it comes to packaging, disease detection, pollutant removal, and nutrition delivery, nanotechnology holds out a lot of promise for improving food safety. It provides chances to raise food standards, lower food waste, and lower the health hazards brought on by foodborne illnesses. To ensure its safe and responsible application in the food business, as with any newly developed technology, comprehensive analysis of the possible dangers and appropriate regulation are needed. We can open the door to a future of food production and consumption that is safer and more sustainable by utilizing the advantages of nanotechnology while tackling the related issues.

NANOMATERIAL APPLICATIONS IN FOOD PROCESSING AND PRESERVATION

Food processing and preservation procedures can be improved with nanomaterials, increasing food quality, safety, and shelf life. Details on how nanoparticles are used in food processing and preservation are provided below:

a) **Advanced Packaging Materials with Improved Barrier Properties, Improved Mechanical Strength, and Increased Durability**: Nanomaterials are used to build advanced packaging materials with Improved Shelf Life and Food Safety. Major applications consist of:

 Nanocomposite Films: To make nanocomposites, packaging films are combined with nanoparticles or nanofillers. These coatings have outstanding gas barrier qualities that stop oxygen, moisture, and other pollutants from permeating. They keep the freshness, flavor, and nutritional value of food products while extending their shelf life.

 Active Packaging: Materials for packaging contain nanomaterials with antibacterial capabilities. As a result, food safety is increased and spoiling is decreased. They also release active ingredients, such as silver nanoparticles, that stop the growth of bacteria, fungi, and other pathogens.

b) **The use of nanoparticles with inherent antibacterial capabilities to control pathogens**: Nanoparticles with antimicrobial qualities include silver, zinc oxide, and titanium dioxide. To reduce infections and enhance food safety, they can be used in a variety of ways:

 Surface Coatings: Surface coatings that are used on objects that come into touch with food, such as utensils and equipment, can comprise nanoparticles. The danger of contracting a foodborne illness is decreased because these coatings stop bacterial growth, limiting cross-contamination.

 Food Sanitizers: In settings where food is processed, nanoparticles can be utilized as sanitizing agents. They have shown efficient against a variety of viruses, fungi, and bacteria, lowering the risk of contamination during processing and handling.

c) **Nanocomposites for Producing Flexible, Lightweight, and Strong Food Containers**: In order to create food packaging materials with better characteristics, nanomaterials are employed.

 Lightweight Materials: Nanocomposites can be designed to be lightweight and have great strength and durability. This characteristic lessens the weight of the packing, the cost of shipping, and the environmental impact.

 Mechanical Strength: Excellent mechanical strength is a property of nanocomposite materials that lowers the possibility of package damage during handling, transportation, and storage.

 Transparency and Flexibility: Nanocomposites can be made to be translucent, flexible, and heat-resistant. These characteristics enable the creation of cutting-edge packaging options, like pouches and films, that safeguard and display food products.

d) **Nanosensors for Real-Time Food Quality and Freshness Monitoring**: Nanosensors enable real-time monitoring of food quality parameters, enabling prompt decision-making and ensuring customer safety:

 Gas Sensors: Nanosensors are able to identify and measure particular gases, like ammonia or volatile organic compounds, released during food rotting. By providing early warning signals of food rotting, these sensors support the preservation of food quality and safety.

 Biosensors: Biosensors based on nanomaterials can identify pollutants, allergies, and pathogens that are present in food. They offer sensitive and quick detection techniques that allow for quick responses and guarantee the security of food goods.

Food processing and preservation using nanomaterials offers chances to increase food quality overall while also enhancing food safety, extending shelf life, reducing waste, and improving food safety. Although there are worries about the potential effects of nanomaterials on human health and

the environment, continued study, risk assessment, and regulatory measures are required to ensure their safe and responsible usage in the food business.

MICROENCAPSULATION OF FOOD INGREDIENTS USING NANOMATERIALS

In the microencapsulation process, tiny droplets or particles of active substances are encased in a barrier, creating microcapsules or nanoparticles. With their distinctive qualities and advantages, nanomaterials play a vital role in microencapsulation. Here are the details on microencapsulation of food ingredients using nanomaterials:

a) **Definition and Benefits of Microencapsulation:** The active elements, such as tastes, nutrients, or bioactive chemicals, are enclosed in a protective matrix in a process known as microencapsulation. Polymers, proteins, lipids, or a combination of these can all be used as encasing materials. The following advantages of microencapsulation in the food industry:

 Protection: The enclosing matrix serves as a barrier, guarding against environmental deterioration brought on by elements like light, oxygen, moisture, or temperature.

 Improved Stability: The greater stability of encapsulated substances ensures that their quality, bioactivity, and sensory properties are retained over an extended length of time.

 Controlled Release: Because microcapsules allow for controlled release, the substances they contain are released gradually and continuously over time. The flavor, aroma, or nutritional value of the constituents in food products may be improved by this controlled release.

 Targeted Delivery: By allowing for the targeted delivery of compounds to particular bodily locations, microencapsulation increases the ingredients' absorption, bioavailability, and effectiveness.

b) **Nanocapsules and nanospheres utilized as carriers in microencapsulation**: Nanocapsules and nanospheres are nanoscale structures. They provide the following benefits:

 Dimensions and Surface Area: The high surface area-to-volume ratio of nanocapsules and nanospheres makes it possible to efficiently encapsulate a lot of active substances.

 Enhanced Protection: The encapsulated substances are shielded from outside influences by the small size of nanocapsules and nanospheres, which offers enhanced protection.

 Customizability: Nanomaterials may be designed to have particular features, such as controlled release traits or focused delivery systems.

 Compatibility: Nanocapsules and nanospheres can be made to work with various food matrices by carefully designing them, allowing for their inclusion into a variety of food products.

c) **Controlled Release of Flavors, Nutrients, and Bioactive Compounds**: By enabling the controlled release of encapsulated substances, microencapsulation offers a number of advantages.

 Flavor Release: Flavors can be encapsulated to allow for a controlled release throughout food intake, extending the sensory experience and enhancing flavor perception.

 Nutrient Preservation: Encapsulation safeguards delicate nutrients from deterioration during processing or storage, such as vitamins, antioxidants, or omega-3 fatty acids.

 Controlled Bioactive Release: Bioactive substances, such probiotics or plant extracts that have health advantages, can be encapsulated and released in a controlled way to ensure their survival and usefulness.

d) **Increased Stability and Targeted Delivery of Encapsulated Ingredients**: Using nanomaterials in microencapsulation offers additional benefits for stability and targeted delivery.

 Enhancing Stability: Nanomaterials can make encapsulated substances more stable by acting as a barrier against outside influences that might otherwise result in deterioration or loss of functionality.

 Targeted Delivery: Nanomaterials enable the design of encapsulation systems with particular release properties, allowing for the targeted delivery of ingredients to selected body regions, such as the gastrointestinal tract or particular tissues.
The food sector has chances to increase ingredient preservation, stability, controlled release, and targeted distribution through the use of microencapsulation employing nanomaterials. Food producers can improve the nutritional value, sensory qualities, and product quality by encapsulating delicate or bioactive substances, creating novel and useful food items.

RELATED FOOD SAFETY ISSUES AND CONCERNS

The use of nanoparticles in the food business has several opportunities, but it also creates food safety concerns. It is critical to address these concerns and guarantee that nanomaterials are used responsibly. Here are the specifics on food safety issues and concerns:

a) Potential Risks and Uncertainties Related to Food Nanomaterials:

 Toxicity: Because of their small size, high surface area, and unique physicochemical features, some nanomaterials may be toxic to humans. To ensure the safety of nanomaterials in food, their potential toxicity must be properly studied.

 Bioaccumulation: There is fear that certain nanomaterials may accumulate in the body and cause long-term health problems if consumed in food.

 Environmental Impact: Nanomaterials released into the environment, such as during food processing or disposal, may have unanticipated ecological repercussions and have an impact on ecosystems.

b) **Nanomaterial Safety Assessment and Regulatory Considerations**:

 Risk Assessment: To evaluate the possible risks and hazards connected with nanoparticles utilized in food, comprehensive safety assessments are required. This includes determining their toxicity, bioavailability, migratory potential, and exposure limits.

 Testing and Standardization: To evaluate the safety and performance of nanomaterials intended for food applications, robust testing methodologies and defined protocols are required. These should include physicochemical characterisation, toxicological testing, and an assessment of environmental effect.

 Regulatory Frameworks: Around the world, regulatory bodies are drafting rules and regulations for nanomaterials in food. The goals of these frameworks are to ensure the safety, labeling, and transparency of nanomaterial use in food items.

c) Consumer Acknowledgement and Transparency Concerning Food Nanotechnology:

 Consumer Perception: Due to concerns about safety, potential dangers, and unfamiliarity with the technology, consumer acceptability of nanoparticles in food products may vary. It is critical to address customer attitudes and give clear and accurate information on the benefits, safety measures, and laws associated with food nanotechnology.

 Labeling and Transparency: It is critical to notify customers about the presence of nanomaterials in food products through labeling transparency. Clear labeling procedures can increase consumer trust, allow for more informed choices, and make traceability easier.

d) **Achieving Responsible Application by Balancing Innovation and Precaution**:

 Precautionary Approach: When dealing with uncertainty and potential dangers linked with nanomaterials, it is critical to follow the precautionary approach. This principle advocated for taking preventive steps even in the absence of total scientific assurance, with public health and safety as the first priority.

 Responsible Innovation: It is critical to strike a balance between innovation and safety. Researchers, industry, and regulatory authorities should work together to progress nanotechnology responsibly, including thorough safety assessments, appropriate rules, and open communication.

It is possible to leverage the benefits of nanoparticles in food while protecting consumer health and guaranteeing responsible industry application by addressing potential concerns, conducting rigorous safety studies, setting clear laws, and fostering openness. Continuous research, risk assessment, and regulatory measures are required to promote nanotechnology while protecting

public safety and increasing consumer confidence.

NANOMATERIALS AND ITS FUTURE PROSPECTS

Nanomaterials have shown considerable promise in transforming the food sector, with numerous advantages for food processing, preservation, microencapsulation, analysis, and safety. These nanomaterials with unique features and capabilities have the potential to improve food quality, lengthen shelf life, and improve safety measures. However, ethical and safe nanomaterial deployment in the food business necessitates ongoing research, risk assessment, and regulatory procedures.

a) **A recap of Nanomaterials' Importance in the Food Industry**:

 Food Processing and Preservation: Nanomaterials allow for the creation of superior packaging materials with increased barrier qualities, antibacterial properties, and controlled release systems. They aid in the preservation of food freshness, the extension of shelf life, and the reduction of spoiling.

 Microencapsulation: Nanomaterials enable the microencapsulation of food ingredients, which protects them from deterioration, improves stability, and enables controlled release. This improves flavor, nutrient retention, and bioactive component delivery.

 Food Analysis and Safety: Nanomaterials enable rapid and sensitive detection of pollutants, pathogens, and food-borne risks. Nanosensors, assays based on nanomaterials, and labeling approaches improve food analysis and provide real-time monitoring for improved food safety.
 Consumer Benefits: Nanomaterials allow for the creation of functional foods with enhanced sensory properties, nutritional value, and customized delivery mechanisms. They provide chances for product variety, innovation, and individualized nutrition.

b) **The need for Continued Research, Risk Assessment, and Regulatory Measures**: Nanomaterials have a lot of potential, but their usage in the food business must be safe and sustainable. The following factors are crucial:

 Continued study: It is important to do ongoing study to better understand how nanomaterials interact with food matrices and the human body as well as their potential hazards and advantages. The creation of secure and efficient applications based on nanomaterials is influenced by this research.

 Risk Assessment: In-depth risk analyses are necessary to examine the potential effects of nanomaterials used in food on human health and the environment. Assessing their toxicity, degrees of exposure, and long-term impacts are all part of this process.

 Regulatory Measures: To ensure the safe usage, labeling, and transparency of nanomaterials in food, comprehensive regulations and guidelines must be implemented. Regulatory organizations are essential in assessing safety, establishing standards, and encouraging ethical innovation.

 Stakeholder Collaboration: To resolve issues, promote transparency, and increase public confidence in the safe and responsible use of nanomaterials in food, collaboration among researchers, industry stakeholders, regulatory authorities, and consumer groups is essential.

Finally, it should be noted that nanomaterials have enormous potential to revolutionize the food business by enhancing food processing, preservation, microencapsulation, analysis, and safety. However, to ensure the protection of consumer health, the environment, and the integrity of the food supply chain, their safe and responsible deployment requires continual research, risk assessment, and regulatory measures. The food sector can take use of nanoparticles' benefits while assuring their safe and sustainable use by striking a balance between innovation and caution.

PRODUCT DEVELOPMENT

NEED FOR PRODUCT DEVELOPMENT

Product development is by definition a future-oriented practice. It is an effort to foresee the future needs of the market place and to translate this information into state-of-the-art products.

FOOD product development involves more than just creating the perfect recipe. Companies must plan extensively, work hard, and research for an extended period of time in order to produce new food products.
Companies engage in new product development with the hopes of gaining new customers, expanding into new geographic markets, increasing profits, elevating brand excitement, or increasing market shares. Companies large and small introduce thousands and thousands of new food products each year. The time spent developing new food products ranges from 6 months to 5 years, depending on the degree of new technology and innovation.

IDEA GENERATION

- Companies use varying techniques to generate ideas for new products. Marketing teams may be charged with the central development of ideas with supplementation of researcher input. Ideas may also come from consumer input.
- After idea generation, the major steps in developing a new food product may be divided into four phases: screening, feasibility, test marketing, and commercialization.

SCREENING After an idea has been created, the steps of product development.

- Screening is the most critical step in a product development project. Thorough testing of product concepts can assist a firm in deciding whether to invest time and money into a venture, or to abandon the efforts completely. Project ideas should be congruent with organizational goals.
- Collaboration of departments during the screening step helps to evaluate individual areas involved in product development including financial and legal considerations, process and equipment availability, purchasing power and ingredient accessibility, shifts in the marketplace, and consumer perceptions.
- Examining markets and conducting consumer research are vital to product screening.
- Consumer testing is essential when screening products. Without consumer testing, companies have no way of knowing consumer needs, desires, and willingness to purchase.

Initial screening may reveal useful information for later marketing schemes.

FEASIBILITY

- Feasibility considerations for a business include regulations, technology, and finances. By setting up an interdepartmental team, the tools will be available to answer initial questions of attainability that may be introduced at any stage during the development process.
- Regulations At the start of a project, firms must be cognizant of the state and/ or federal agencies that regulate a product.

Technology

- In order to launch a new food product, the necessary equipment, facilities, and processes needed to manufacture a product must be established. When products are found to not be technologically feasible, the project should be terminated.

Facilities

- The facility that is available for processing should be considered. If the company has acquired a new plant, the water supply, equipments needed, machineries, man power, sewage systems and all other required materials will need to be inspected.
- The conditions inside the plant, such as temperature and relative humidity control, should be taken into account. Facilities in areas with high humidity and heat in the summer without controls for these conditions may have to modify operating conditions to produce quality foods.

Finances

- Before a food product is created for sale, an understanding of all Feasibility production and marketing costs is required. A detailed cost analysis should be made prior to manufacture.
- The two types of costs to consider include fixed costs and variable costs. Annual fixed costs are those that will not change in any one year, regardless of the level of production. These costs include equipment, building, property taxes, and other items that do no fluctuate due to changes in production.
- Variable costs are expenditures that vary with the volume of production, such as hired labor, raw ingredients, packaging materials, fuel, electricity, utilities, and other items used during production. Variable costs should be carefully examined prior to test marketing and commercialization to implement a unit price in order for the new product to make a profit.

Shelf Life

- Shelf life is the determination of how long a product will hold its quality as perceived by customers. The shelf life of a product is important when considering distribution channels.
- Shelf life can be determined through the use of accelerated or real time testing. Safety New product developers should consider the safety risks of their products. History of outbreaks and published safety risks of certain product categories can help give clues on risk factors of which processors should be cognizant. For example, peanut butter producers must use controls to test for possible salmonella contamination after a large outbreak in early 2009 caused over 400 people to become ill and at least five deaths.
- Some products are susceptible to the growth of spoilage and pathogenic microorganisms.

Allergens and physical contaminants, like metal shavings from processing equipment, can pose safety threats to consumers as well.

FORMULATION OF THE PRODUCT

Varying ingredients, processing parameters, and packaging options will be utilized in order to find the best combination to create the desired product. Sound statistical analysis and good record keeping are critical at this step.

All formulas and experiments should be detailed in a laboratory notebook. Each entry should include all necessary details.

- **Test kitchen**

  Research is carried out, to formulate a number of recipes and specify the ingredients to be used. Several versions of the product are usually made, using slightly different ingredients or processes, i.e. the products are prototyped in the company's test kitchen, often by a professional chef or food consultant.

  A small number of staff, experienced in sensory evaluation, test the products and evaluate them informally. On some occasions, if a company does not have a test kitchen, an initial concept for a product is sent to an independent developer.

- **Sensory Evaluation**

It is usual for sensory evaluation to be carried out at many stages of the development process.

Trained assessors comment on the appearance, odour, taste and texture of the products to make sure that the product being developed displays the desired characteristics.

The assessors are able to do this with great accuracy**.**

- **Pilot plant**

  When an acceptable product has been made in the kitchen it is then produced using pilot scale equipment. A pilot plant is a small version of the equipment used in manufacture. Many manufacturers have these facilities.

  They produce the product on a small scale, but in the same way as it would happen during large scale manufacture. This allows them to judge whether the product can be successfully produced on a large scale, and whether it meets the original specification.

  Enough product must be made for market research and sensory evaluation to be undertaken.

- **Product specification**

  After testing and modifying the product on the pilot plant, a final product is created. This details exact ingredients and precise methods of production.

  The specification is very important as it will be used for the production of each batch of the product to ensure consistency.

- **TEST MARKETING**

  Many manufacturers test new products on potential consumers before moving on to large scale production.

  If the new product is targeted at a particular group of the population, the consumers will be selected from that group , e.g. vegetarians. Otherwise, the company will try to ensure that different sub-groups are represented, e.g. males and females from a variety of backgrounds across a range of ages.

  Three main methods are used:

  Group discussions – usually up to 8 people are involved in a discussion run by a trained person. The aim is to hear a detailed view of a new product.

  Street interviews – people are asked to taste samples of the product and say what they like or dislike ; this produces a 'snapshot' of opinions about the test product.

  Home trials - consumers are provided with a product and asked to try it at home. This gives the opportunity to sample the product in a 'real life' situation and to have a response from the whole family. The trial may result in minor changes to the product specification and the manufacturing process.

- **Scale up**

  Once the company is confident that the product is likely to succeed, it is ready for large scale manufacture to begin.

  The manufacturing process is sometimes organised in unit operations, such as size reduction, mixing and cooking. These are controlled to maintain consistent product quality, safeguard, staff health, food safety standards and to avoid problems that may stop the production line running, which would result in 'down-time'.

- **Packaging**

  Packaging is an important part of a consumer's appeal for a product, especially with first time purchases.

  It is important to consider how consumers will view the packaging and if it will convey the product's quality goals, such as being a high quality premium product or a generic grade.

  Marketing, product developers, and packaging engineers should consider the types of packaging materials that are being used on competitors' products and how to set themselves apart.

- **Launch of the new product**

  Advertising then takes place to let the consumer know that the product is available. Finally the new product is launched

- **Distribution**

  Products that require special distribution needs include frozen and refrigerated foods. Organizations should consider the cost of special distribution.

  Other distribution considerations include the radius in which the product will be available. Will the product have nationwide or regional distribution? The distribution radius can also influence the packaging needs.

PRODUCT LIFE CYCLES

- Products go through cycles during the duration of their sales. During the introductory period when the product is first launched, companies heavily promote their products in order to attract customers.
- In-store demonstrations are sometimes used to attract customers that may not try the product otherwise. Discounts and coupons can help spike sales of a new product as well.
- In this introductory period, the costs on the company are high and the returns are minimal.
- The next phase in the cycle is a strong growth period. At this time, repeat buyers may decide to purchase the product on a regular basis. Word of mouth from customers may begin to attract other new customers.
- Expansion to new markets may assist in growing sales. Costs continue to be high, but profits are improving. The next phase is a decline in the growth rate. Repeat buyers decline, new markets have been tapped out, the competition begins to grow, and there are new costs associated with trying to attract attention to the product. Profits are still good in this phase.
- The stability period sees no growth in sales due to consumer fatigue. There is little excitement about the product, and sales stagnate. Costs and profits break even, but profits may begin to decline.
- In the product decline phase, competitive products begin to beat out the product, and promotions are too costly to be beneficial. Sales of the product decline, and the product becomes costly to maintain. The product is unprofitable. At this point, companies must decide if it is necessary to cease manufacturing the product.

SHELF LIFE

Foods are perishable by nature. Numerous changes take place in foods during processing and storage. It is well known that conditions used to process and store foods may adversely influence the quality attributes in foods.

Due to Storage for a certain period, one or more quality attributes of a food may reach an undesirable state. At that instant, the food is considered unsuitable for consumption and it is said to have reached the end of its shelf life. Shelf life of a food product may be actually defined as the time between the production and packaging of the product and the point at which it becomes unacceptable under defined environmental conditions. Storage and distribution are necessary links in the food chain, and hence considered as factors influencing the shelf life.

Major modes of food deterioration

During storage and distribution, foods are exposed to a wide range of environmental conditions. Environmental factors such as temperature, humidity, oxygen and light can trigger several reaction mechanisms that may lead to food degradation. As a consequence of these mechanisms, food may be altered to such an extent that they are either rejected by the consumer, or they may become harmful to the person consuming them.

Chemical, physical and microbiological changes are the leading causes of the food deterioration. A discussion on these changes follows.

Physical Changes

Physical changes are caused by mishandling of foods during harvesting, processing and distribution; these changes lead to reduced shelf life of foods. Crushing of dried snack during distribution seriously affects their quality. Dried foods when kept in high humidity may pickup moisture and become soggy.

Chemical Changes

During the processing and storage of foods, several chemical changes occur that involve the internal food components and the external environmental factors. These changes may cause food deterioration and reduce the shelf life.

The most important chemical changes are associated with enzymic action, oxidative reactions, particularly lipid oxidation that alters the flavor of many lipid containing foods, and non-enzymic browning that causes changes in appearance.

Fruits upon cutting tend to brown rapidly at room temperature due to the reaction of phenolase with the cell constituents that are released upon cutting of the tissue in presence of oxygen. Enzymes such as lipoxygenase, if not denatured during the blanching process, can influence food quality even at sub-freezing temperatures. In addition to the temperature, other environmental factors such as oxygen, water and pH induce deleterious changes in foods that are catalyzed by enzymes.

The presence of oil and fats containing unsaturated fatty acids is a prime reason for the development of rancidity in foods during storage as long as oxygen is available. Development

of off-flavours which is markedly noticeable in rancid foods is the result of autoxidation of unsaturated fatty acids. The generation of free radicals during the autocatalytic process leads to other undesirable reactions, for example, loss of vitamins, alteration of colour, and degradation of proteins.
In addition to lipid oxidation, there are other chemical reactions that are induced by light such as loss of vitamins, and browning of meats. Non-enzymic browning is a major cause of quality change and degradation of nutritional content in many foods. This type of browning reaction occurs due to the interaction between reducing sugars and amino acids. These reactions result in the loss of protein solubility, darkening of lightly coloured dry products and the development of bitter flavours. Environmental factors such as temperature, water activity and pH have an influence on non-enzymic browning.

Microbiological changes

Microbes have the ability to multiply at high rates when favourable conditions are present. Prior to harvest, fruits and vegetables have generally good defense mechanisms against microbial attacks, however, after separation from the plant they can easily succumb to microbial proliferation. Similarly, meat upon slaughter is unable to resist rapidly growing microbes. Microbial growth in foods results in food spoilage with the development of undesirable sensory characteristics and in certain cases the food may become unsafe for consumption. The pathogenicity of certain microorganisms is a major safety concern in processing and handling of foods upon ingestion. Microorganisms such as Salmonella species and Escherichia coli strains cause infection while others such as Aspergillus flavus, Clostridium botulinum and Staphylococcus aureus produce chemicals in foods that are toxic to humans.

Evaluation of food quality

A common practice employed to evaluate the shelf life of a given food product is to determine changes in selected quality characteristics over a period of time. One may consider quality of a food as a gross measure of the food deterioration occurring in food item. However, it should be recognized that the term quality is meant to encompass 35 several quality attributes or characteristics. From a consumer's standpoint, the sensory expectations derived from the presence (or absence) of desirable (or undesirable) characteristics of a given food determine the quality of a product. Therefore a food product noted for its high quality has more of the desirable characteristics.

Procedures for determination and monitoring of shelf life
Direct shelf life determination requires batches of samples to be taken at significant stages in the development or modification of the product. These samples should be examined during storage, usually under controlled environmental conditions, until their quality becomes unacceptable. The time when this occurs is the maximum product shelf life, and therefore the determination necessarily requires at least this time to complete.

Significant sampling stages within the programme of shelf life evaluation include:

1. The successful experimental kitchen or pilot plant batch. At this stage it is possible to investigate formulation, process or packaging changes to improve the shelf life without the costs of factory time and material quantities. 2. The successful full scale factory batch. This is the most important sampling stage. This will provide the data for the setting of shelf life and specification standards. 3. The first continuous production trial. Examination of products should confirm the data from earlier samplings

Shelf life samples should be subjected to conditions effectively simulating the normal storage and distribution conditions the food is likely to encounter.

Shelf life examination is done by employing appropriate methods of: **sensory evaluation; chemical analysis and microbiological analysis.**
In sensory evaluation appearance, smell, texture and flavour being the main attributes to assess. Such assessments are frequently inexact as there may not be a suitable control sample with which to compare the stored samples, this being particularly so for new products. However, under appropriate test conditions using control sample it is possible to get a fair idea about the quality and acceptability of stored products.

Quantitative measurements, for example of colour, texture, viscosity and amount of water or oil separation should be included if they either closely relate to the sensory quality or can be used as reliable indicators of quality deterioration. In addition to subjective assessments, other tests may be necessary. These may include tin content of products in unlacquered cans, vitamin content where a claim is made. Microbiological examination of fresh and stored products is highly essential in order to determine whether they are safe for human consumption.